CS-548: Cloud-native Software Architectures

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Website: https://www.csd.uoc.gr/~hy548/

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Assignment 4: Extending

1. (*20% grade*) Provide the YAML that allows you to manage a custom resource of type "Fruit" with Kubernetes. An example instance is the following:

   ```
   apiVersion: hy548.csd.uoc.gr/v1
   kind: Fruit
   metadata:
     name: apple
   spec:
     origin: Krousonas
     count: 3
     grams: 980
   ```

   Also, provide the kubectl commands needed to:
   a. Install the custom resource.
   b. Create the above instance.
   c. Return the new instance in YAML format.
   d. Return a list of all available instances.

2. (*40% grade*) The example code for CRDs using the Python Kopf library is available on GitHub (https://github.com/chazapis/hy548, in the `crds-kopf` folder). Extend the example so that:
   a. The executable `controller.py` runs inside a container. Provide the `Dockerfile`. Build and upload the new container to Docker Hub.
   b. Provide `greeting-controller.yaml`, which will create a Deployment with the container you made. Make sure the necessary permissions are set so the controller can read the "Greeting" CRDs and create the corresponding Deployments with the `hello-kubernetes` container in all namespaces.

   Provide the commands you used to verify that the deployment works correctly.

3. (*40% grade*) The example code for the webhooks is available on GitHub (https://github.com/chazapis/hy548, in the `webhooks` folder). Extend the example so that:
   a. The executable `controller.py` runs inside a container. You will need to modify the code so that it uses SSL. Provide the `Dockerfile`. Build and upload the new container to Docker Hub.
   b. The `webhook.yaml` should use a service/deployment based on the new container instead of the Nginx proxy. Provide the new `webhook.yaml`.

   Provide the commands you used to verify the webhook works correctly.

Notes:

- The assignment is personal.
- All exercises contribute equally to the overall grade (unless individual percentages are defined).
- Ask for questions and clarifications at the beginning of lectures or via email.
- Write down your answers in a Markdown-formatted text file in either Greek or English.
- Place all files in a folder created for the assignment. Submit your answers in the private GitHub repository that you will use for the course. Share the repository with the instructor (username "chazapis").
- Commits done after the exercise's deadline will be ignored.